

$$\hat{u}(x, t) = \underbrace{g(x) e^{-\frac{1}{2}(15t)^2}}_{\text{known IC, fading out}} + \underbrace{\tanh^2(25t)}_{\text{gate}} \cdot \underbrace{N_\theta(x, t)}_{\text{learned "NN term"}}$$

= $g(x)$ at $t=0$;
 gone by $t \approx 0.2$

0 at $t=0$ with zero slope
 (the square kills the
 slope); open by $t \approx 0.1$

the blank in the template
 (PINNs and final/ML
 FNO both fill it)

Both initial conditions hold exactly, by construction: $\hat{u}(x, 0) = g(x)$ and $\hat{u}_t(x, 0) = 0$